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Optimization of solid-state fermentation process of diets and its effects on growth performance and intestinal function in growing pigs

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Abstract

Background Biological fermentation can improve animal growth performance and meat quality by optimizing feed nutritional properties. However, the complex fermentation parameters require further systematic optimization. This study aimed to establish and optimize a solid-state fermentation (SSF) process, evaluate changes in the feed microbial community and flavor metabolites, and investigate their effects on muscle development and intestinal barrier function in growing pigs. Here, we developed a synergistic SSF strategy for pig feed using combinations of 4 probiotics and 11 enzymatic preparations, 16S rDNA-seq and flavoromics-seq were employed to investigate the dynamic changes in microbial communities and flavor compounds post-fermentation. Subsequently, 32 Duroc × Jinfen White pigs were fed diets containing 10% SSF to assess growth performance, intestinal health and muscle development.

Results The optimal fermentation ratio of *Bacillus subtilis*, *Lactobacillus plantarum*, *Saccharomyces cerevisiae* and *Aspergillus niger* is 1:2:3:3, with a temperature of 36 °C, an inoculation rate of 93%, a moisture content of 72%, and a time of 4.1 days. SSF significantly enhanced the nutritional value of feed by increasing the ash, organic matter (OM), crude protein (CP), and ether extract (EE) content, while simultaneously reducing the concentration of anti-nutritional factors. Sequencing identified 17 differential microbes and 116 flavor compounds, with the relative abundance of *Lactobacillus* and the *Firmicutes* significantly increased, 2-octenal and vanillin imparting sweet, fruity, and grassy notes to the feed. Meanwhile, RNA-seq revealed 320 DEGs in muscle tissue following fermented feed supplementation, which are mainly enriched in pathways related to cytochrome P450 drug metabolism and arginine biosynthesis. Additionally, H&E staining results indicated that fermentation significantly increased the villus height, crypt depth, and villus-to-crypt ratio in the small intestine of growing pigs, and the levels of tight junction proteins Claudin-1 and ZO-1

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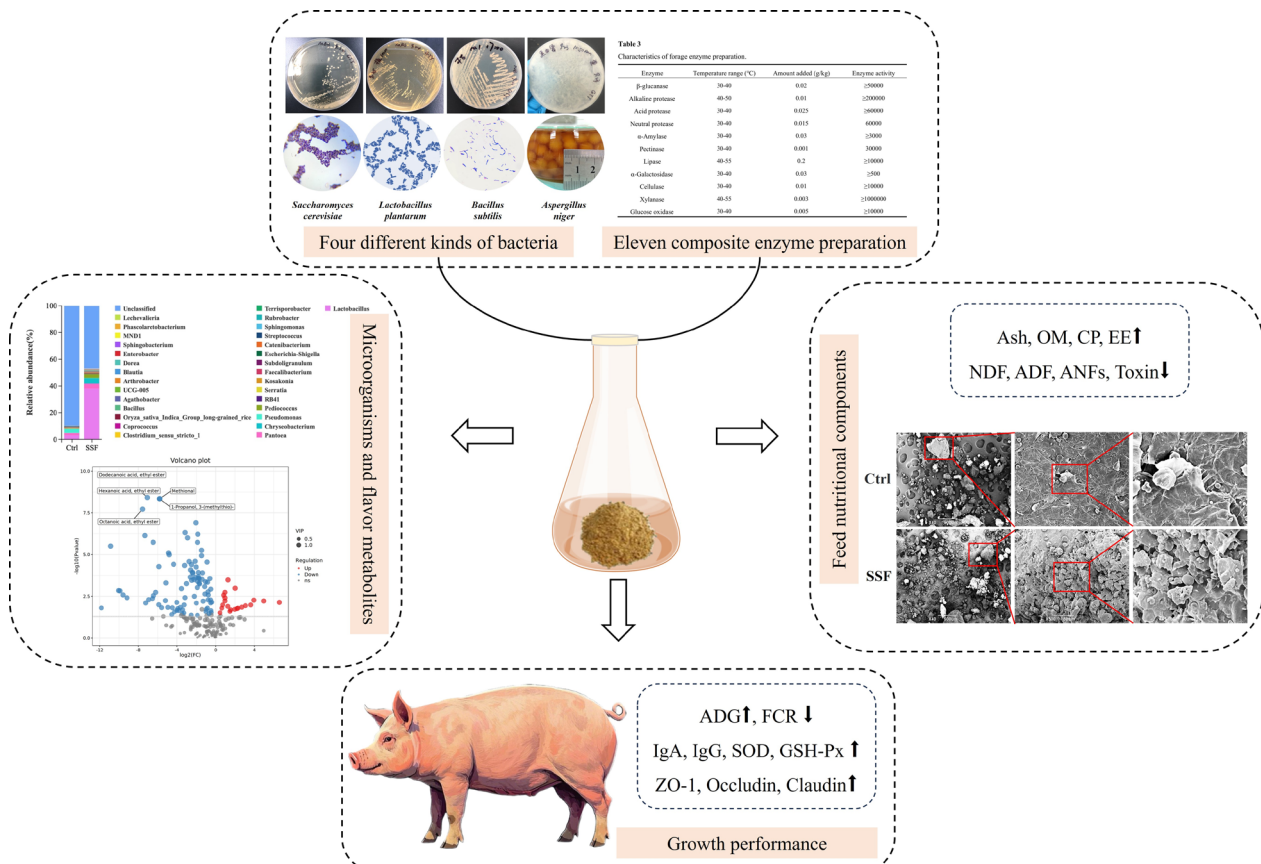


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in the jejunum were significantly higher than those in the ctrl group. Subsequent correlation analysis indicated that *Firmicutes* may influence pig growth performance and IL-6, TNF- α levels by affecting their metabolites.

Conclusion Our findings establish and optimize an SSF process that markedly elevates feed nutritional value, enriches beneficial microbes, and fosters the production of unique flavor metabolites. When fed to growing pigs, it effectively enhances growth performance and antioxidant capacity while improving small-intestinal morphology and barrier function.

Graphical Abstract



Keywords Microbial-enzyme synergy, Feed microbiota, Feed flavoromics, Muscle development, Intestinal barrier

Background

Corn and soybean meal, as high-quality nutritional sources in pig diets, provide essential energy, protein [1]. However, it is limited by anti-nutritional factors such as soybean antigenic proteins, trypsin inhibitors (TI), and non-starch polysaccharides (NSP), which negatively affect feed utilization, nutrient release and absorption in pigs, and may lead to intestinal allergies and diarrhea in pigs [1, 2]. Microbial fermentation can effectively eliminate anti-nutritional factors and certain macromolecular substances in feed. Previous studies have predominantly focused on liquid-state fermented feed [3], while solid-state fermentation (SSF) has gradually emerged as the preferred process in modern feed industry, owing to its

comprehensive advantages in enhanced nutritional bio-availability, improved product stability, and reduced environmental footprint [4–6]. SSF is a type of feed produced through microbial fermentation on a solid substrate, has been demonstrated to reduce energy consumption in feed production through multiple mechanisms, such as eliminating the need for substantial water inputs during fermentation and minimizing energy requirements for raw material processing [7]. Nonetheless, the fermentation process involves multiple conditions, and the substances that play a key role in enhancing feed quality remain unclear. Therefore, establishing and optimizing SSF processes is of great significance.

Compared with microbial SSF alone, microbial-enzyme co-fermentation can more degrade and utilize substrates, significantly reducing the anti-nutritional factors (ANFs) in feed and enhancing the value of feed or raw materials [8, 9]. For instance, during the fermentation process, the addition of cellulase and lactic acid bacteria can effectively reduce ANFs in soybean meal and enhance its nutritional value [10, 11]. Concurrently, the addition of non-starch polysaccharide enzymes to fermented feed can increase the diversity of the pig gut microbiota, particularly enhancing the relative abundance of beneficial bacteria such as *Bifidobacterium* [12]. Moreover, the addition of enzymes can compensate for the insufficiency of enzyme production by the microorganisms, accelerating substrate degradation. Research has found that enzymes can influence the fermentation efficiency of microorganisms by affecting the production of lactic acid and changes in pH levels, and the co-fermented feed is rich in probiotics and enzymatic metabolic products, which are beneficial for maintaining animal gut health [13]. It has also been reported that microbial-enzyme co-fermentation can break down high-molecular-weight proteins into various peptides [14].

Research indicates that fermentation is a dynamic process during which the composition of the microbial community undergoes significant changes. For instance, Tang effectively reduced the relative abundance of pathogenic bacteria such as *Escherichia-Shigella* in the intestines of finishing pigs using SSF, while increasing the relative abundance of beneficial bacteria like *Firmicutes* and *Clostridium* [3]. There have also been experimental findings that SSF feed can increase the abundance of beneficial bacteria, such as *Lactobacillus*, thereby enhancing the balance of the intestinal microbiota and immune function [15]. Simultaneously, fermentation can also significantly improve the volatile flavor compounds in feed [16]. It has been reported that during the solid-state fermentation process, enzymes produced by microorganisms can break down complex organic matter in feed, such as cellulose, hemicellulose, and pectin, releasing a greater amount of flavor compounds and nutrients, thereby enhancing the nutritional value of the feed [17]. However, research on the microbial community structure and changes in flavor compounds within fermented feed is still relatively scarce. Therefore, this study aims to optimize the strain ratios and fermentation conditions in SSF using a four-strain mixture, analyze changes in microbial community structure and flavor metabolites in fermented feed, and investigate their effects on growth performance and intestinal function. This research will provide a scientific basis for the optimization of SSF processes and a comprehensive evaluation of its nutritional value.

Table 1 Orthogonal test factors and levels design table of L9 (3⁴) for strains proportion

Level	Factors			
	(A) <i>B. subtilis</i>	(B) <i>L. plantarum</i>	(C) <i>S. cerevisiae</i>	(D) <i>A. niger</i>
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3

Table 2 Design of RSM experimental factors and levels

Factor	Levels		
	-1	0	1
A Inoculum quantity(%)	6	9	12
B Moisture content(%)	60	80	100
C Time(d)	3	4	5

RSM= Response Surface Methodology;

Materials and methods

Establishment of solid-state fermented (SSF) feed process

The *Lactobacillus plantarum* (*L. Plantarum*, BNCC336421) and *Bacillus subtilis* (*B. Subtilis*, BNCC109047) used in the experiment were purchased from Bena Culture Collection, while *Saccharomyces cerevisiae* (*S. Cerevisiae*, CICC1355) and *Aspergillus niger* (*A. niger*, CICC40273) were obtained from the China Center of Industrial Culture Collection. According to the orthogonal experimental design (Table 1), the activated four strains were inoculated into 300 mL conical flasks containing 50 g of feed. The fermentation was conducted for 72 h under the following conditions: an initial inoculation amount of 3%, a temperature of 36 °C, a moisture content of 80%, and the natural pH. The strain ratios were optimized by assessing the contents of trichloroacetic acid-soluble protein (TCA-SP), crude fiber (CF), and the β -conglycinin. Subsequently, the conditions were optimized based on the selected strain ratio. Specifically, single-factor and response surface methodology (RSM) experiments (Table 2) were conducted to investigate the effects of fermentation times (1 day, 2 days, 3 days, 4 days, 5 days, and 6 days), fermentation temperatures (24 °C, 28 °C, 32 °C, 36 °C, and 40 °C), water content (60%, 80%, 100%, 120%, and 140%), and microbial inoculation amount (6%, 9%, 12%, 15%, and 18%) on the TCA-SP and CF. After the compound enzyme preparation was fully dissolved in distilled water at the corresponding dosage specified in Table 3, it was uniformly mixed with the feed. Subsequently, the SSF feed was produced via the bacteria-enzyme synergistic fermentation process under optimized strain ratio and fermentation conditions. The nutritional composition and levels of the basal diet are shown in Table 4. All experiments were conducted with three biological replicates.

Table 3 Characteristics of forage enzyme preparation

Enzyme	Temperature range (°C)	Amount added (g/kg)	Enzyme activity
β-glucanase	30–40	0.02	≥ 50,000
Alkaline protease	40–50	0.01	≥ 200,000
Acid protease	30–40	0.025	≥ 60,000
Neutral protease	30–40	0.015	60,000
α-Amylase	30–40	0.03	≥ 3000
Pectinase	30–40	0.001	30,000
Lipase	40–55	0.2	≥ 10,000
α-Galactosidase	30–40	0.03	≥ 500
Cellulase	30–40	0.01	≥ 10,000
Xylanase	40–55	0.003	≥ 1,000,000
Glucose oxidase	30–40	0.005	≥ 10,000

Table 4 Nutrient composition and nutrient level of basal diet

Ingredient	Contents (%)
Corn	67.45
Whey powder	3.00
Soybean oil	1.23
Soybean	20.61
Fish meal	2.00
Lysine hydrochloride	0.55
Met(98%)	0.23
Thr(98%)	0.23
Trp(98%)	0.08
Limestone	0.48
CaHPO ₄	1.57
NaCl	0.56
Premix ²⁾	2.00
Total	100.00
Nutrient levels ¹⁾	Contents (%)
DM	92.30
Moisture	7.70
CP	18.36
EE	4.10
Ash	6.87
ADF	7.38
NDF	15.88
Ca	0.69
Total Phosphorus	0.65
Lys	1.20
Met + Cys	0.71
Thr	0.76
Trp	0.24

Met=Methionine; Thr=Threonine; Trp=Tryptophan; DM=Dry Matter; CP=Crude Protein; EE=Ether extract; ADF=Acid detergent fiber; NDF=Neutral detergent fiber; Lys=Lysine; Cys=Cystine

¹⁾ Moisture, CP, EE, Ash, NDF and ADF were measured values, the others were calculated values

²⁾ Premix provided per kilogram of diet: Vitamin A 10,000 IU; Vitamin D 3500 IU; Vitamin E 70 IU; Vitamin K3 3 mg; Vitamin B1 3 mg; Vitamin B2 10 mg; Vitamin B6 6 mg; biotin 0.4 mg; pantothenate acid 30 mg; niacin 30 mg; Cu 110 mg (as copper sulfate); Mn 10 mg (as manganese sulfate); Fe 140 mg (as ferrous sulfate); I 0.5 mg (as potassium iodide); Se 0.3 mg (as sodium selenite)

Determination of feed nutritional value

The SSF feed was taken out from the incubator, ground into fine powder after being dried in an oven at 65 °C for over 48 h, and then stored sealed at 4 °C for later use. Following the methods of previous studies [18, 19], the feed was analyzed for crude protein (CP), ether extract (EE), ash, and neutral detergent fiber (NDF), while also determining the content of acid detergent fiber (ADF) and trichloroacetic acid-soluble protein (TCA-SP). The contents of soybean globulin (YJ820239), trypsin inhibitor (TI, YJ820238), β-conglycinin (YJ831129), phytohemagglutinin (PHA, YJ854229), deoxynivalenol (DON, ml103502), zearalenone (ZEN, ml036116), aflatoxin B1 (AFB1, ml036115), and T-2 toxin (TS, ml036120) were determined according to the manufacturer's instructions (Shanghai Enzyme-Linked Biotechnology Co., Ltd., China).

Determination of small peptides and analysis by electron microscopy

To preserve the native structure of the feed particles, they were carefully picked up with tweezers and laid flat on the sample stage without applying pressure, ensuring the integrity of the original morphology for scanning electron microscope observation across multiple viewing fields. Additionally, add the denatured feed to a protein gel for electrophoresis, followed by staining and destaining with Coomassie brilliant blue dye.

16S rDNA sequencing

16S rDNA sequencing was performed on the basal feed and SSF feed. After DNA extraction from the samples, a library was constructed and sequenced on the Illumina PE250 platform. The DADA2 software was used for data assembly, read filtering, redundancy removal, denoising, and chimera elimination. The ASV sequences were aligned against the SILVA or UNITE database (Table S1), and taxonomic classification was annotated using the RDP classifier software. The annotation results were visualized using KRONA, and sequence counts at the phylum and genus levels were tallied. Alpha diversity indices, including observed_species (Sob), Chao1, Shannon, and Pielou, were analyzed using QIIME (version 1.9.1). Beta diversity was assessed using the weighted UniFrac algorithm to analyze differences among microbial communities in different groups, with results displayed in a Principal Coordinates Analysis (PCoA) plot. Venn diagram analysis was conducted using the vegan package in R, followed by intergroup difference analysis using the LEfSe software. Functional annotation and community function prediction were performed using the Tax4Fun software. The OTU abundance table was combined with the “species-gene” network to output the

relative functional abundance for each sample (NCBI: PRJNA1275368).

Flavor metabolomics analysis

The 1 g sample was mixed with 10 μ L of internal standard solution and incubated at 60 °C for 30 min. The Solid-Phase Microextraction (SPME) fiber was aged at 270 °C for 10 min, then placed in the incubation chamber. After adsorption at 60 °C for 30 min, the SPME fiber was inserted into the injection port (desorbed at 250 °C for 5 min) and then aged again at 270 °C for 10 min. The injection volume was 10 μ L, with an injection temperature set at 250 °C and a constant helium flow rate of 1.0 mL/min. Detection was performed using the GC $\times$ GC-TOFMS chromatography system (LECO, St. Joseph, MI, USA). The scan range was set at m/z 35–550, with the transfer line and ion source temperature maintained at 250 °C, an acquisition rate of 200 spectra/second, an electron impact energy of 70 eV, and a detector voltage of 2019 V. Continuous scan MS data were collected to generate total ion current (TIC) chromatograms. Prior to multivariate statistical analysis, all data were subjected to standardized preprocessing. Principal component analysis (PCA) was first applied to classify samples and assess intergroup trends, thereby excluding poorly reproducible and outlier samples. Orthogonal partial least squares-discriminant analysis (OPLS-DA) was then employed to identify metabolites with a variable importance in projection (VIP) greater than 1 as potential biomarkers. Statistically significant differential metabolites were further screened by combining univariate statistical methods (t-test or one-way ANOVA) with the criteria of p -value < 0.05 and VIP > 1. Additionally, based on the sensory attributes of flavor compounds from the FlavorDB database, an association network was constructed using the igraph package to elucidate intrinsic relationships among flavor metabolites (NGDC: OMIX011743).

RNA-seq

In this study, total RNA was extracted from the rapidly frozen *longissimus dorsi* muscle using the Trizol method and assessed for purity with the NanoDrop ND-2000 (Thermo Fisher Scientific, Wilmington, DE) and validated for integrity with the Agilent 2100 Bioanalyzer. Stranded mRNA libraries were constructed using the Illumina TruSeq Stranded mRNA LT Kit and sequenced on the NovaSeq 6000 platform with 150 bp paired-end reads. Raw data underwent quality control (ctrl) with fastp to filter out low-quality data, yielding clean reads. These reads were then aligned to the reference genome using HISAT 2. Gene expression levels were quantified using Stringtie to reconstruct transcripts and RSEM to calculate expression levels across all genes in each sample. Differentially expressed genes (DEGs) were identified

based on criteria of $FDR < 0.05$ and $|\log_2 \text{FoldChange}| \geq 1$. Subsequently, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis was conducted using the online platform I-Sanger (NCBI: PRJNA1282860).

Experimental animal management

This study selected 48 Duroc $\times$ Jinfen White pigs with a body weight of 17.87 ± 2.83 kg at 60 ± 3 days of age. The pigs were randomly divided into two groups, each with 4 replicates per group and 6 pigs per replicate (an equal number of males and females): the Ctrl group and the SSF group. The Ctrl group was fed a basal feed, while the SSF group received the experimental diet composed of 10% SSF feed mixed with 90% basal diet. After a 5-day adaptation period, a formal feeding trial was conducted for 28 days. During the trial, the pigs had ad libitum access to feed and water. The environmental conditions were controlled at a temperature of 21.2 ± 0.85 °C and a relative humidity of $65.8 \pm 4.19\%$. Before the end of the trial, the pigs were fasted for 12 h, then euthanized by electrocution [20, 21] and sample was collection.

Measurement of growth performance and sample collection

Individuals participating in the experiment were weighed at the beginning and end of the trial, and daily feed intake per replicate was recorded. These data were used to calculate average daily gain (ADG), average daily feed intake (ADFI), and feed conversion ratio (FCR). Six pigs were randomly selected from each group to collect blood and tissue samples. After euthanasia, blood was collected via anterior vena cava puncture into plain vacuum coagulation tubes, and serum was separated by centrifugation. The *longissimus dorsi* muscle was collected from the left side of the carcass, between the 13th and 16th lumbar vertebrae, after slaughter, for the measurement of meat color, shear force, drip loss, and pH value. Simultaneously, collected tissue sample from the *longissimus dorsi*, duodenum, jejunum, ileum, and mid-colon were divided into two portions. One portion was gently rinsed in PBS and subsequently fixed in 4% paraformaldehyde fixative, while the remaining portion was immediately snap-frozen in liquid nitrogen and stored at -80 °C for subsequent analysis.

Immune biochemistry and enzyme-linked immunosorbent assay (ELISA)

Conventional nutritional indicators such as total protein, albumin, globulin, blood glucose, and urea nitrogen were measured using an automatic biochemical analyzer (IDEXX, USA). ELISA kits for detecting interleukin-4 (IL-4, YJ31752), interleukin-6 (IL-6, YJ35656), interleukin-10 (IL-10, YJ31974), and tumor necrosis factor- α (TNF- α , YJ31476) in serum were purchased

Table 5 Effect of 4 bacteria mixed fermentation on TCA-SP

Level	Factors				TCA-SP (%)			
	B. Subtilis (A)	L. Plantarum (B)	S. Cerevisiae (C)	A. Niger (D)	y1	y2	y3	y
1	1	1	1	1	3.352	3.467	3.434	3.418
2	1	2	2	2	3.566	3.648	3.691	3.635
3	1	3	3	3	3.615	3.702	3.609	3.642
4	2	1	2	3	3.533	3.407	3.516	3.485
5	2	2	3	1	3.380	3.981	3.363	3.575
6	2	3	1	2	3.101	3.270	3.095	3.155
7	3	1	3	2	3.210	3.232	3.451	3.298
8	3	2	1	3	3.374	3.495	3.024	3.298
9	3	3	2	1	3.254	3.320	3.298	3.290
K1	32.085	30.603	29.613	30.849				
K2	30.647	31.522	31.232	30.264				
K3	29.657	30.264	31.544	31.276				
k1	3.565	3.400	3.290	3.428				
k2	3.405	3.502	3.470	3.363				
k3	3.295	3.363	3.505	3.475				
R	0.270	0.140	0.214	0.112				
Optimal combination	A ₁ B ₂ C ₃ D ₃							

B. Subtilis: *Bacillus subtilis*; *L. Plantarum*: *Lactobacillus plantarum*; *S. Cerevisiae*: *Saccharomyces cerevisiae*; *A. Niger*: *Aspergillus niger*; TCA-SP: trichloroacetic acid-soluble protein

Table 6 Table of variance analysis of orthogonal test

Variation source		SS	df	MS	F	P-value	Significance
TCA-SP	B. Subtilis	0.331	2	0.166	6.556	0.007	**
	L. Plantarum	0.094	2	0.047	1.863	0.184	ns
	S. Cerevisiae	0.239	2	0.119	4.723	0.022	*
	A. Niger	0.057	2	0.029	1.135	0.344	ns
	SSe2	0.455	18	0.025			
	Total	1.176	26	0.045			

B. Subtilis: *Bacillus subtilis*; *L. Plantarum*: *Lactobacillus plantarum*; *S. Cerevisiae*: *Saccharomyces cerevisiae*; *A. Niger*: *Aspergillus niger*; SS: Sum of Squares; df: Degree of freedom; MS: Mean Square; F: F-value; SSe2: Sum of Squares for Error

To compare the statistical significance between two groups: ns (not significant) indicates $P > 0.05$, * $P < 0.05$ indicates a significant difference, and ** $P < 0.01$ indicates a highly significant difference

from Shanghai Enzyme-Linked. Malondialdehyde (MDA, A003-1-2), catalase (CAT, A007-1-1), total antioxidant capacity (T-AOC, A015-2-1), superoxide dismutase (SOD, A001-3-2), and glutathione peroxidase (GSH-PX, A005-1-2) were measured using biochemical assay kits from Nanjing Jiancheng.

Histological evaluation

The fixed intestinal tissues were dehydrated using an ASP-200 (Leica, Germany). Tissue sections with a thickness of 5 μm were obtained using a rotary microtome (Leica, Germany), following a previously reported protocol [22]. The tissue sections were stained with hematoxylin-eosin, and images were captured using the EVOSFL Auto Cell Imaging System software. Chorion length, crypt depth, and chorion height/crypt depth ratios were calculated using Image-Pro Plus 6.0 software.

Statistical analysis

In this experiment, all data except for sequencing data were processed using SPSS20.0. An independent samples T-test was used to analyze the data, and the results were expressed as “mean ± SEM”. Statistical significance was indicated by a p-value of less than 0.05.

Results

The effect of strain ratios on solid-state fermentation

The optimal mixing ratios of the four strains were determined through orthogonal experiments. The results showed that the best combination for increasing the TCA-SP value was A₁B₂C₃D₃ (Table 5), with significant effects from *B. subtilis* ($P < 0.01$) and *S. cerevisiae* ($P < 0.05$) on TCA-SP (Table 6). Meanwhile, the best combination for CF degradation was A₁B₃C₃D₃ (Table 7), with significant effects from *S. cerevisiae* ($P < 0.01$) and *B. subtilis* ($P < 0.05$) on CF (Table 8). Additionally, the two mixed strains significantly reduced the levels of soybean

Table 7 Effect of 4 bacteria mixed fermentation on CF

Level	Factors				CF(%)			
	B. Subtilis (A)	L. Plantarum (B)	S. Cerevisiae (C)	A. Niger (D)	y1	y2	y3	y
1	1	1	1	1	3.691	3.835	3.564	3.697
2	1	2	2	2	3.746	3.802	3.866	3.805
3	1	3	3	3	3.458	3.265	3.301	3.341
4	2	1	2	3	3.871	4.062	3.821	3.918
5	2	2	3	1	3.826	3.741	3.659	3.742
6	2	3	1	2	3.905	3.982	3.785	3.891
7	3	1	3	2	3.848	3.513	3.582	3.648
8	3	2	1	3	3.854	3.611	3.837	3.767
9	3	3	2	1	3.980	3.785	3.931	3.899
K1	32.529	33.788	34.066	34.014				
K2	34.654	33.943	34.866	34.030				
K3	33.942	33.394	32.193	33.081				
k1	3.614	3.754	3.785	3.779				
k2	3.850	3.771	3.874	3.781				
k3	3.771	3.710	3.577	3.676				
R	0.236	0.061	0.297	0.105				
Optimal combination	A1 B3 C3 D3							

B. Subtilis: *Bacillus subtilis*; L. Plantarum: *Lactobacillus plantarum*; S. Cerevisiae: *Saccharomyces cerevisiae*; A. Niger: *Aspergillus niger*

Table 8 Table of variance analysis of orthogonal test

Variation source	SS	df	MS	F	P-value	Significance
CF						
A-B. Subtilis	0.260	2.000	0.130	9.319	0.0017	*
B-L. Plantarum	0.018	2.000	0.009	0.638	0.5396	ns
C-S. Cerevisiae	0.418	2.000	0.209	14.993	0.0001	**
D-A. Niger	0.066	2.000	0.033	2.354	0.1236	ns
SSe2	0.251	18.000	0.014			
Total	1.013	26.000	0.039			

To compare the statistical significance between two groups: ns (not significant) indicates $P > 0.05$, * $P < 0.05$ indicates a significant difference, and ** $P < 0.01$ indicates a highly significant difference

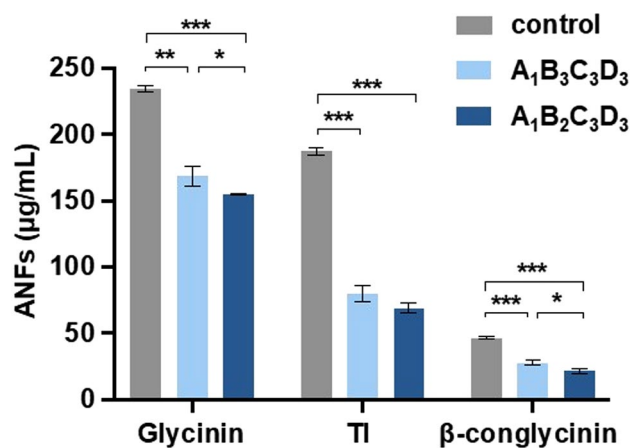


Fig. 1 The effects of different mixed-strain combinations and fermentation conditions on solid-state fermented feed and anti-nutritional factors. Statistically significant differences between two groups were considered: * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$. Data are expressed as means $\pm$ SEM ($n = 4$)

globulin, TI, and β -conglycinin in the feed ($P < 0.001$), with the degradation effects of the $A_1B_2C_3D_3$ group being significantly higher than that of the $A_1B_3C_3D_3$ group (Fig. 1). Therefore, the $A_1B_2C_3D_3$ combination was selected as the optimal mixed strain ratio for further screening.

The effects of different fermentation conditions on solid-state fermented feed

Single-factor and response surface optimization experiments were further conducted to verify the effects of moisture content, inoculation amount, fermentation time, and temperature on SSF feed. As shown in Fig. 2A-D, the highest levels of TCA-SP were observed at a moisture content of 80%, while CF levels were at their lowest, indicating that 80% is the optimal moisture content. Similarly, comparative analysis of TCA-SP and CF responses across experimental conditions determined the optimum inoculum size at 9%, fermentation time at 4 days, and fermentation temperature at 36 °C. The response surface optimization experiment indicated that the established

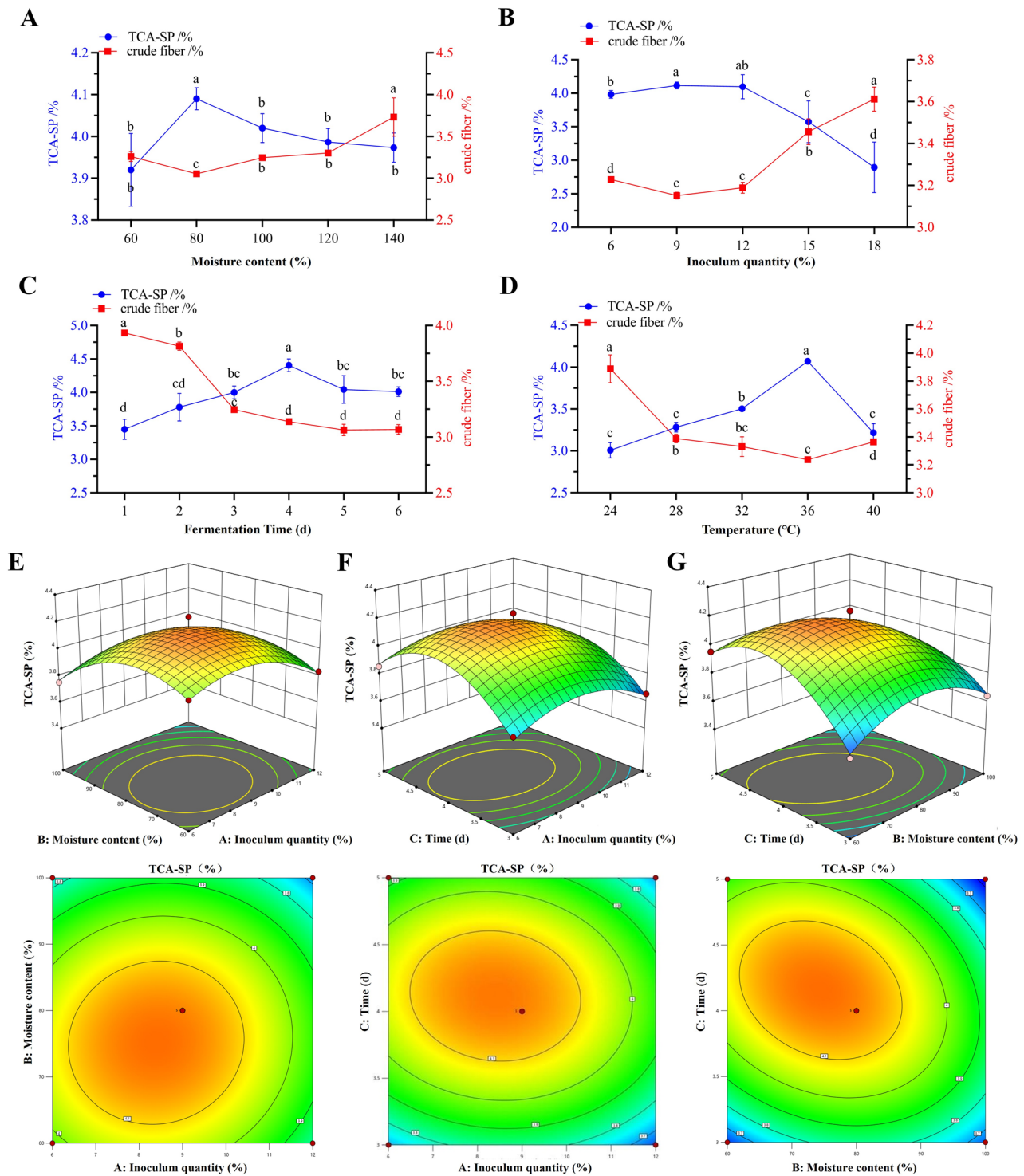


Fig. 2 Effects of different fermentation conditions on solid-state fermented feed and response surface optimization experiments. The effect of **A** Moisture content, **B** Inoculum quantity, **C** Fermentation time, **D** Temperature on solid-state fermented feed. **E** Response surface plots and contour plots of inoculum quantity and moisture content. **F** Response surface plots and contour plots of moisture content and fermentation time. **G** Response surface plots and contour plots of inoculum quantity and fermentation time. ^{a, b, c} Different superscripts in each row for each factor differ significantly ($P < 0.05$)

Table 9 Results of response surface variance analysis

Source	Sum of Squares	df	Mean Square	F-value	P-value	Significance
Model	0.7758	9	0.0862	21.29	0.003	**
A-Inoculation amount	0.0281	1	0.0281	6.94	0.0337	*
B-Moisture content	0.062	1	0.062	15.3	0.0058	**
C-Fermentation time	0.0253	1	0.0253	6.25	0.041	*
AB	0.0011	1	0.0011	0.2609	0.6252	
AC	0.003	1	0.003	0.7337	0.4201	
BC	0.0406	1	0.0406	10.03	0.0158	*
A ²	0.1026	1	0.1026	25.33	0.0015	**
B ²	0.1489	1	0.1489	36.79	0.0005	**
C ²	0.3037	1	0.3037	75.02	< 0.0001	**
Residual	0.0283	7	0.004			
Lack of Fit	0.0037	3	0.0012	0.1979	0.8929	ns
Pure Error	0.0247	4	0.0062			
Total	0.8041	16				

To compare the statistical significance between two groups: ns (not significant) indicates $P > 0.05$, * $P < 0.05$ indicates a significant difference, and ** $P < 0.01$ indicates a highly significant difference

Table 10 Variance analysis of quadratic regression equation

Parameter	Value
Std. Dev.	0.0636
Mean	3.87
C.V. %	1.65
R ²	0.9648
Adjusted R ²	0.9194
Predicted R ²	0.8792

second-order regression model was highly significant ($P < 0.01$, Table 9). The analysis of variance showed that the model had an R² value of 0.96 and an adjusted R² of 0.92 (Table 10). By using response surface optimization software to fit the data, the resulting equation was obtained as $Y = 4.15 - 0.0592A - 0.088B + 0.0563C + 0.0162AB - 0.0237AC - 0.1008BC - 0.1561A^2 - 0.1881B^2 - 0.2686C^2$. Furthermore, based on the interactions among moisture content, fermentation time, and inoculation amount, the optimal TCA-SP value was predicted. The predicted conditions were an inoculation amount of 9.283%, moisture content of 71.539%, and fermentation time of 4.112 days. Under these conditions, the TCA-SP value in the solid-state mixed-strain fermented feed was 4.112% (Fig. 2E-G).

The impact of solid-state fermented feed with strain-enzyme on feed

To address the issue of insufficient enzyme production during mixed-strain fermentation, microbial-enzyme synergistic fermentation was employed for optimization. The results showed that, compared with the A₁B₂C₃D₃ group, the A₁B₂C₃D₃+complex enzyme group had significantly lower crude fiber content ($P < 0.05$) and significantly higher acid-soluble protein content ($P < 0.05$), indicating better performance of microbial-enzyme synergistic fermentation (Fig. 3A). Meanwhile, after SSE,

the boundaries of the feed cell walls became indistinct (Fig. 3B), and the proteins were degraded from large molecules to below 25 kDa after 4 days of fermentation (Fig. 3C). Furthermore, the nutritional components in the feed were analyzed. The results showed that the SSF group had significantly higher contents of CP, CF, and ash compared to the ctrl group ($P < 0.01$), while the contents of NDF and ADF were significantly lower than those in the ctrl group ($P < 0.01$) (Table 11). Additionally, compared to the ctrl group, the contents of ANF and DON and ZEN were significantly reduced ($P < 0.01$) (Tables 12 and 13), indicating that feed fermentation more effectively reduced the risk of vomiting and poisoning by toxins.

The impact of solid-state fermentation on the microbial community structure of feed

The impact of microbial-enzyme synergistic fermentation on feed microbial composition was investigated using 16S rDNA sequencing. The results showed that the alpha diversity in the SSF group was significantly higher (Fig. 4A), and the microbial community structures between the two groups were distinctly separated (Fig. 4B). Meanwhile, the Venn diagram indicated that the ctrl group had 2,574 ASVs, while the SSF group had 4,117 ASVs (Fig. 4C). At the phylum level, the proportions of *Cyanobacteria* and *Proteobacteria* were significantly reduced in the SSF group ($P < 0.01$), while the proportions of *Firmicutes*, *Acidobacteriota*, and *Gemmatimonadota* were significantly increased ($P < 0.01$) (Fig. 4D). At the genus level, the core genera in the ctrl group were *Pseudomonas* and *Lactobacillus*, while the core genus in the SSF group was *Lactobacillus* ($P < 0.01$). The SSF group also significantly increased the proportions of *Lactobacillus*, *Pediococcus*, and *Acidobacterium RB41*, while significantly reducing the proportion of *Streptococcus* (Fig. 4E).

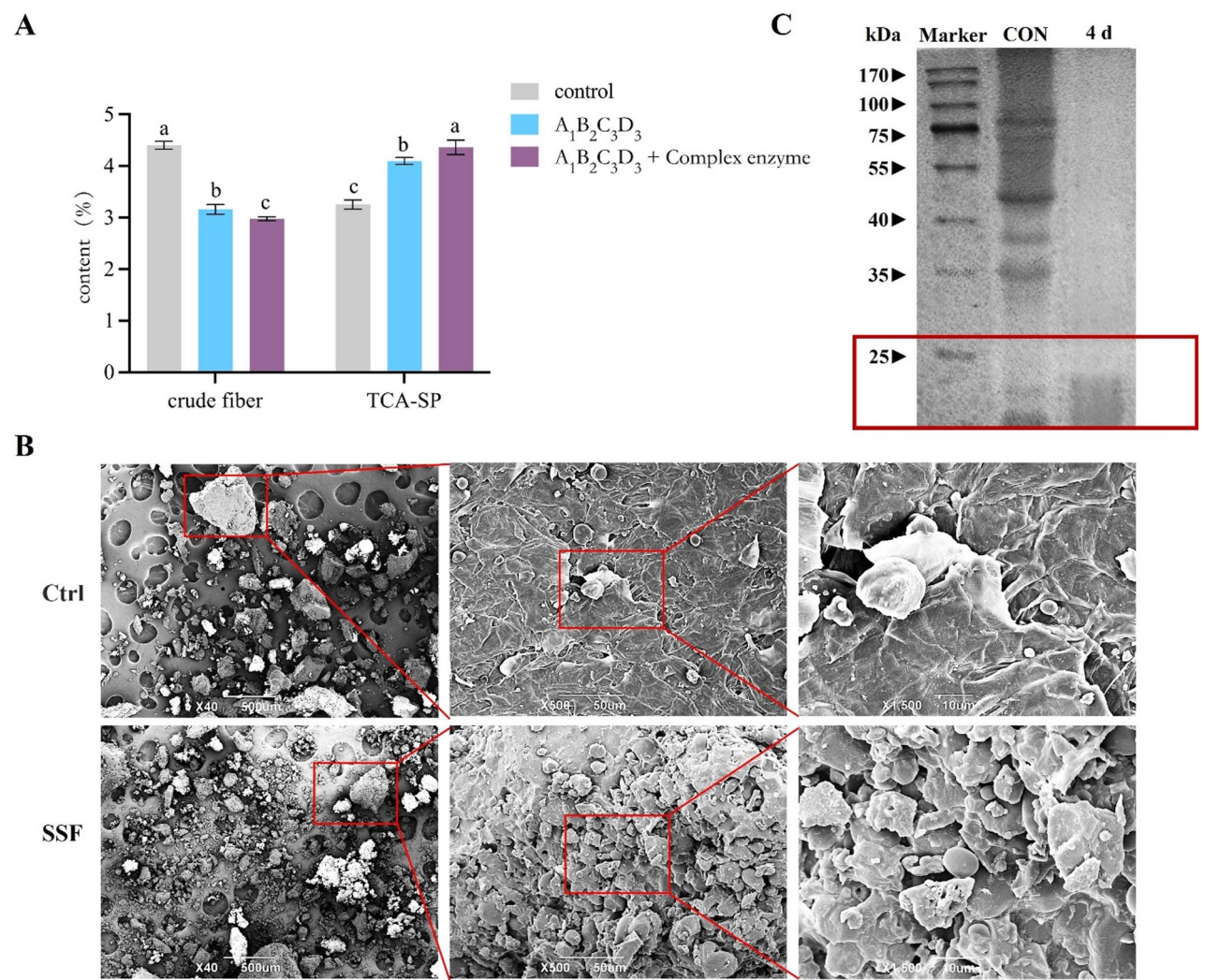


Fig. 3 Surface structure of solid-state fermented feed and changes in protein molecular weight by SDS-PAGE. **A** The synergistic fermentation effect of strains and enzymes. **B** Effect of SSF on feed surface structure (40×, 500×, 1500×). **C** SDS-PAGE of SSF feed at fermentation times of 4 d. ^{a, b, c} Different superscripts in each row for each factor differ significantly ($P < 0.05$)

Table 11 Quality of SSF feed (as air-dry basis)		
Items	Ctrl	SSF
Water, %	7.70±0.01	6.10±0.13**
Ash, %	6.83±0.04	7.35±0.12**
OM, %	85.47±0.04	86.54±0.17**
CP, %	18.36±0.07	22.86±0.18**
EE, %	4.10±0.04	4.89±0.34**
NDF, %	7.38±0.39	4.86±0.10**
ADF, %	15.88±0.91	11.42±0.63**

OM: Organic matter; CP: Crude protein; EE: Ether extract; NDF: Neutral detergent fiber; ADF: Acid detergent fiber

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means±SEM ($n = 4$)

Table 12 Effects of SSF feed on content of anti-nutritional factors		
Anti-nutritional factor	Ctrl	SSF
α-conglycinin (μg/g)	24.94±1.20	14.80±0.58**
β-conglycinin (μg/g)	48.79±1.94	32.85±0.53**
Globulin (μg/g)	331.05±14.10	212.99±9.98**
Soybean trypsin inhibitor (μmol/L)	69.43±5.05	46.56±4.45**
PHA	599.59±34.87	203.03±12.05**

PHA: phytohemagglutinin

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means±SEM ($n = 4$)

LEfSe analysis of intergroup differences revealed that the ctrl group had 10 specific microbial genera, while the SSF group had 17 differentially abundant bacterial communities, with *Lactobacillus* and *Pediococcus* being the main specific genera (Fig. 4F). KEGG functional

Table 13 Effects of SSF on content of toxin in feed

Toxin(μg/kg)	Safety standards(μg/kg)	Ctrl	SSF
DON	< 1000	375.67 ± 22.19	287 ± 10.54**
ZEN	< 500	140.33 ± 6.66	91.33 ± 7.23**
AFB1	< 10	1.6 ± 0.1	< 1.5
TS	< 500	< 50	< 50

DON: Deoxynivalenol; ZEN: Zearalenone; AFB1: Aflatoxin B1; TS: T-2 Toxin
For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means ± SEM ($n = 4$)

prediction showed that the main enrichments were in pathways such as amino acids and nucleotide sugars (Fig. 4G-H).

The impact of solid-state fermentation on feed flavor metabolites

The three-dimensional chromatograms of the fermented feed samples identified by GC × GC-TOFMS are shown in Fig. 5A. PLS-DA analysis revealed that the volatile compounds in each group of feed clustered together, while the two groups were distinctly separated, indicating reliable data (Fig. 5B). The Venn diagram showed that 404 specific compounds were identified in the Ctrl group and 513 in the SSF group (Fig. 5C). The relative contents of aldehyde, acid, ester, and phenolic flavor compounds in the feed increased after fermentation (Fig. 5D).

ROAV analysis indicated that the relative contents of 2-octenal and vanillin in the SSF group significantly increased (Fig. 5E), which also enhanced sweet, fruity, and fatty flavors (Fig. 5F). The volcano plot results showed that the SSF group significantly downregulated several flavor compounds, including ethyl dodecanoate, methionine, ethyl hexanoate, ethyl octanoate, and

3-(methylthio) propanal (Fig. 5G). Finally, to explore the relationship between differential flavor compounds and sensory flavor characteristics, a network was constructed based on the top 10 important sensory characteristics. It was found that flavor metabolites associated with more than five sensory characteristics included ethyl octanoate, 3-nonanone, benzaldehyde, and acetic acid (Fig. 5H).

Effect of solid-state fermented on apparent digestibility and meat quality

Compared to the ctrl group, the SSF group significantly enhanced the ADG in pigs ($P < 0.05$) and reduced the FCR ($P < 0.01$) (Table 14). Concurrently, the relative abundance of *Firmicutes* increased significantly in the SSF group, which was positively correlated with the relative abundance of SCFAs and related metabolic products such as 2,3-Butanediol and 2-Nonenal, while the relative abundance of *Cyanobacteria* was lower in the SSF groups with higher final weight (Fig. 6A-B). Serum analysis revealed that ALB and blood GLU levels in the SSF group were highly significantly elevated compared to the ctrl group ($P < 0.01$), while GLOB and Crea were significantly reduced ($P < 0.05$, Table 15). Immune markers showed that IgA ($P < 0.01$) and IgG ($P < 0.05$) levels in the serum of pigs fed SSF were significantly higher than those in the ctrl group, with highly significant reductions in inflammatory cytokines IL-4 and IL-6 ($P < 0.01$). In terms of antioxidant indicators, SOD and GSH-PX were significantly higher in the SSF group ($P < 0.05$), while MDA was significantly lower ($P < 0.05$, Table 16). Meat quality analysis indicated that the shear force of the *longissimus dorsi* muscle in the SSF group was highly significantly lower than that in the ctrl group ($P < 0.01$, Table 17).

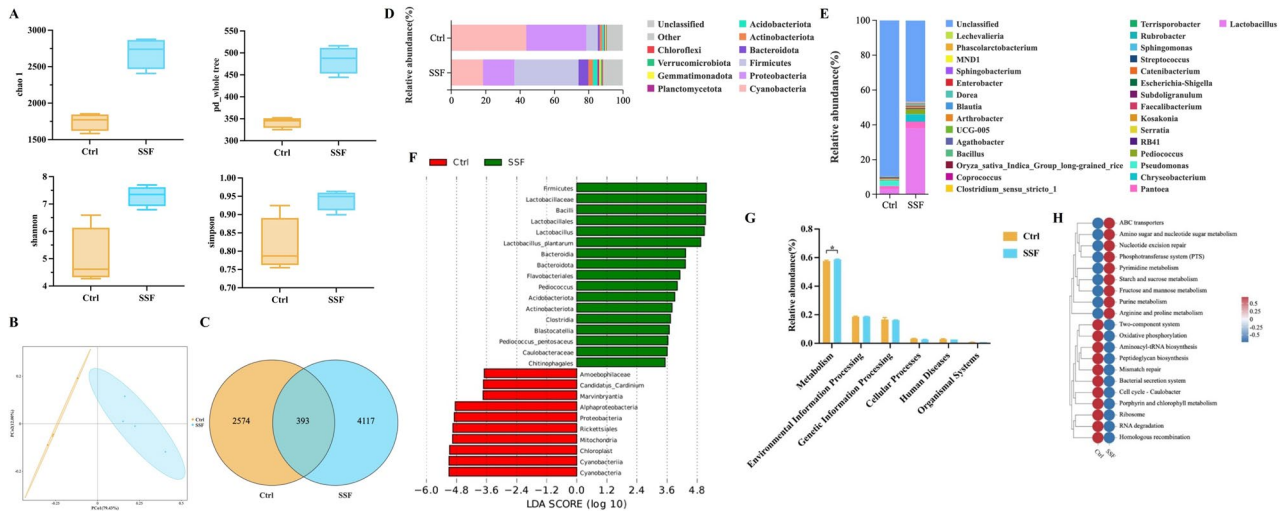


Fig. 4 The diversity of feed microbiota community structure and the relative abundance at the phylum and genus levels. **A** chao index, pd_{whole} index, shannon index, simpson index. **B** The analysis of principal coordinates analysis (PCoA). **C** Venn diagram. Composition of SSF feed at phylum **D** and genus **E** level. **F** The result of LDA discriminant (LDA > 3.5). **G** is the KEGG 1-level pathway. **H** is the 3-level pathway

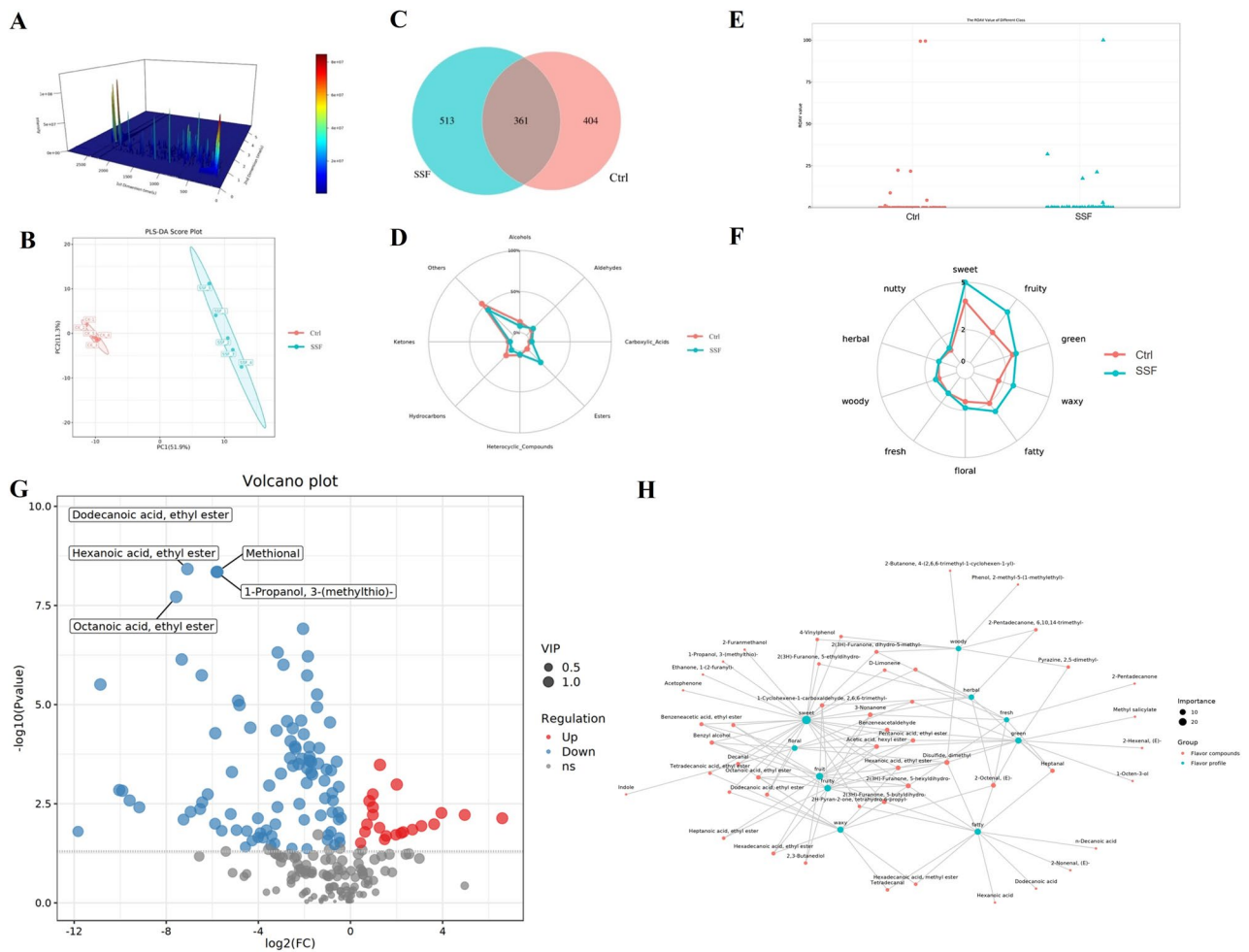


Fig. 5 Changes in Feed Flavor Metabolites. **A** Chromatographic 3D plots. **B** PLS-DA. **C** Venn. **D** Radar map of relative content of flavor substances. **E** ROVA and **F** Sensory Flavor radar map. **G** Volcano Plot of the flavor metabolite. **H** Graph of the association network between sensory flavor features and flavor substances

Table 14 Effects of SSF feed on growth performance of growing pigs

Items	Ctrl	SSF
First weight (kg)	17.27 ± 1.40	18.25 ± 0.40
Final weight (kg)	36.08 ± 1.56	38.65 ± 0.56*
ADG (g)	672.02 ± 41.36	728.50 ± 9.72*
ADFI (g)	1370.70 ± 110.03	1239.28 ± 51.73
FCR	2.01 ± 0.09	1.70 ± 0.06**

ADG: Average daily gain; ADFI: Average daily feed intake; FCR: Feed conversion ratio

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means ± SEM ($n = 4$)

Effect of solid-state fermented feed on pig intestine

HE staining revealed that the villi in the small intestine of the SSF group were more neatly arranged, with jejunal villi exhibiting more leaf-like shapes compared to the finger-like shapes in the ctrl group (Fig. 7A). This change in villus morphology increased the surface area for nutrient absorption. Additionally, the villus height and crypt

depth in all intestinal segments of the SSF group were significantly higher than those in the Ctrl group ($P < 0.01$). The villus-to-crypt ratio in the jejunum and duodenum of the SSF group was also significantly higher than that of the Ctrl group ($P < 0.01$), indicating that feeding SSF helped maintain the integrity of the small intestinal tissue structure in growing pigs (Table 18). Moreover, the expression levels of tight junction proteins Claudin-1, Occludin, and ZO-1 in the jejunal tissue of the SSF group were significantly higher than those in the ctrl group ($P < 0.05$), suggesting that feeding solid-state fermented feed positively impacted the intestinal barrier function in growing pigs (Fig. 7B).

Effect of microbial enzyme co - fermentation feed on the muscle transcriptome of growing pigs

Further investigation was conducted to explore the impact of SSF feed on muscle development in growing pigs. The results showed that the SSF group significantly



For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means $\pm$ SEM ($n = 4$)

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means $\pm$ SEM ($n = 4$)

as the metabolism of xenobiotics by cytochrome P450, drug metabolism-cytochrome P450, and arginine biosynthesis (Fig. 8E). The correlation analysis indicates that an increased abundance of Firmicutes significantly enhances the accumulation of specific fatty acids such as tetradecanoic acid ethyl ester and hexadecanoic acid ethyl ester, potentially directly driving an increase in ADG and a decrease in FCR. Concurrently, the increase in its related metabolites derived from Firmicutes, such

Table 17 Effects of probiotic fermented liquid feed on meat quality

Items		Ctrl	SSF
Drip force (%)		39.67 ± 8.48	31.76 ± 8.89**
Drip loss (%)		8.65 ± 1.99	8.56 ± 1.24
PH	pH _{45min}	6.38 ± 0.24	6.44 ± 0.15
	pH _{24h}	5.48 ± 0.22	5.41 ± 0.11
Meat color lightness (L*)	1 h	43.49 ± 1.58	43.07 ± 1.99
	24 h	50.45 ± 1.65	49.56 ± 3.27
Meat color redness (a*)	1 h	4.92 ± 0.41	5.05 ± 0.37
	24 h	6.80 ± 0.96	7.13 ± 1.41
Meat color yellowness (b*)	1 h	11.55 ± 0.88	12.04 ± 0.83
	24 h	13.73 ± 0.80	14.12 ± 0.58

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means ± SEM ($n = 4$)

as 2,3-butanediol and 2-nonenal, significantly suppresses the expression of pro-inflammatory factors TNF- α and IL-6, reducing systemic inflammation (Fig. 9A-B).

Discussion

SSF is widely used in bio-feed technology due to its broad range of applicable microbial strains, large fermentation substrate volume, and simple operating procedures. However, the SSF process is influenced by microbial types, substrates, and various fermentation conditions, leading to unstable microbial performance and low enzyme production [16, 23, 24]. Some studies have shown that the addition of extra enzymes in combination with microbial fermentation can overcome this challenge and help reduce feed quality losses caused by fermentation

Table 18 Effects of SSF feed on small intestinal histomorphology of growing pigs

Small intestine	Item	Ctrl	SSF
Duodenum	VH(μ m)	535.87 ± 30.66	843.51 ± 59.76**
	CD(μ m)	362.08 ± 11.17	246.26 ± 12.65**
	VH / CD	1.48 ± 0.10	3.429 ± 0.25**
Jejunum	VH(μ m)	323.38 ± 25.82	620.23 ± 35.03**
	CD(μ m)	418.05 ± 21.51	248.05 ± 8.33**
	VH / CD	0.78 ± 0.09	2.50 ± 0.20**

VH: Villus height; CD: Crypt depth

For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $p < 0.05$ and ** $p < 0.01$. Data are expressed as means ± SEM ($n = 4$)

[8–25]. Therefore, this study aims to optimize the fermentation ratios and conditions of *B. subtilis*, *L. plantarum*, *S. cerevisiae*, and *A. niger* by using TCA-SP and CF as key indicators [26] during the SSF process, and to conduct a combined microbial and enzymatic fermentation with 11 enzyme preparations. The results showed that the optimal ratio of (*B. subtilis*: *L. plantarum*: *S. cerevisiae*: *A. niger*) was 1:2:3:3, with a fermentation temperature of 36°C, inoculation amount of 9.3%, moisture content of 72%, and fermentation time of 4.1 days. Meanwhile, the SSF group exhibited significant increases in CP, EE, CF, and NDF content, and high-molecular-weight proteins into smaller peptide molecules. It has been reported that the increase in CP may be associated with the conversion of large molecular substances into more TCA-SP, while the reduction in NDF is beneficial for the absorption of nutrients [27, 28]. Furthermore, ANFs and toxins contained in feed ingredients such as corn and soybean meal

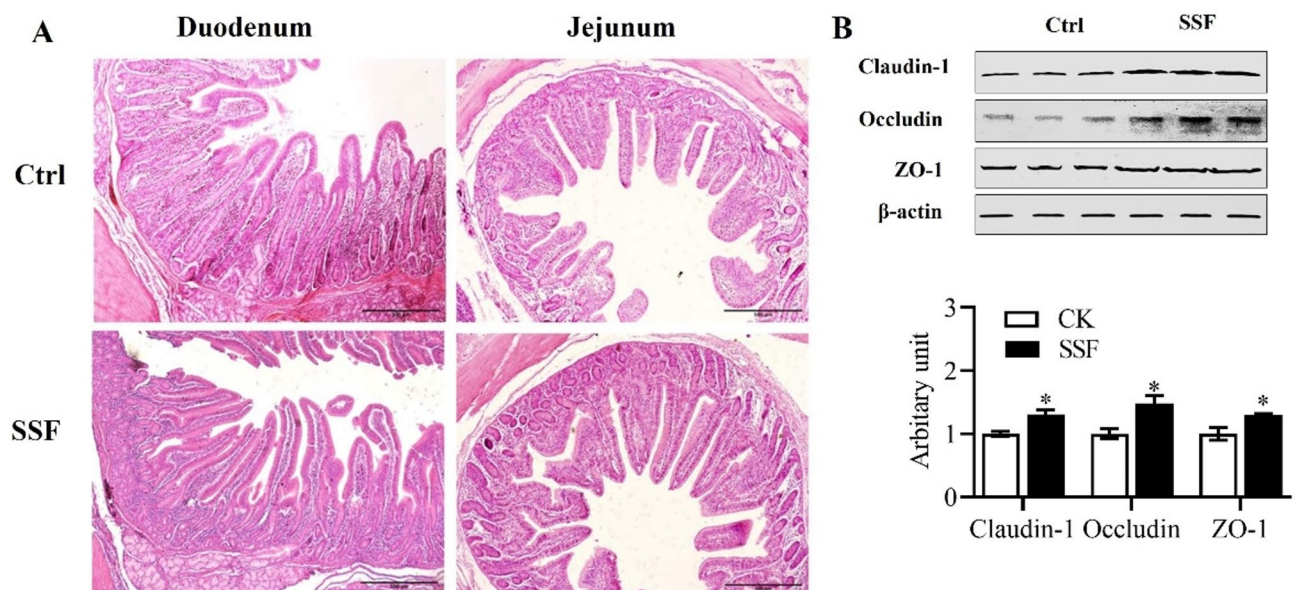


Fig. 7 The impact of strain-enzyme solid-state fermented feed on the intestinal barrier function of growing pigs. **A** Hematoxylin and eosin (H&E) staining of the duodenum and jejunum (100 $\times$). **B** Expression of tight junction factors and proteins in the jejunum of growing pigs in the SSF group. For the same row, statistically significant differences between Ctrl and SSF groups were considered: * $P < 0.05$ and ** $P < 0.01$. Data are expressed as means ± SEM ($n = 4$)

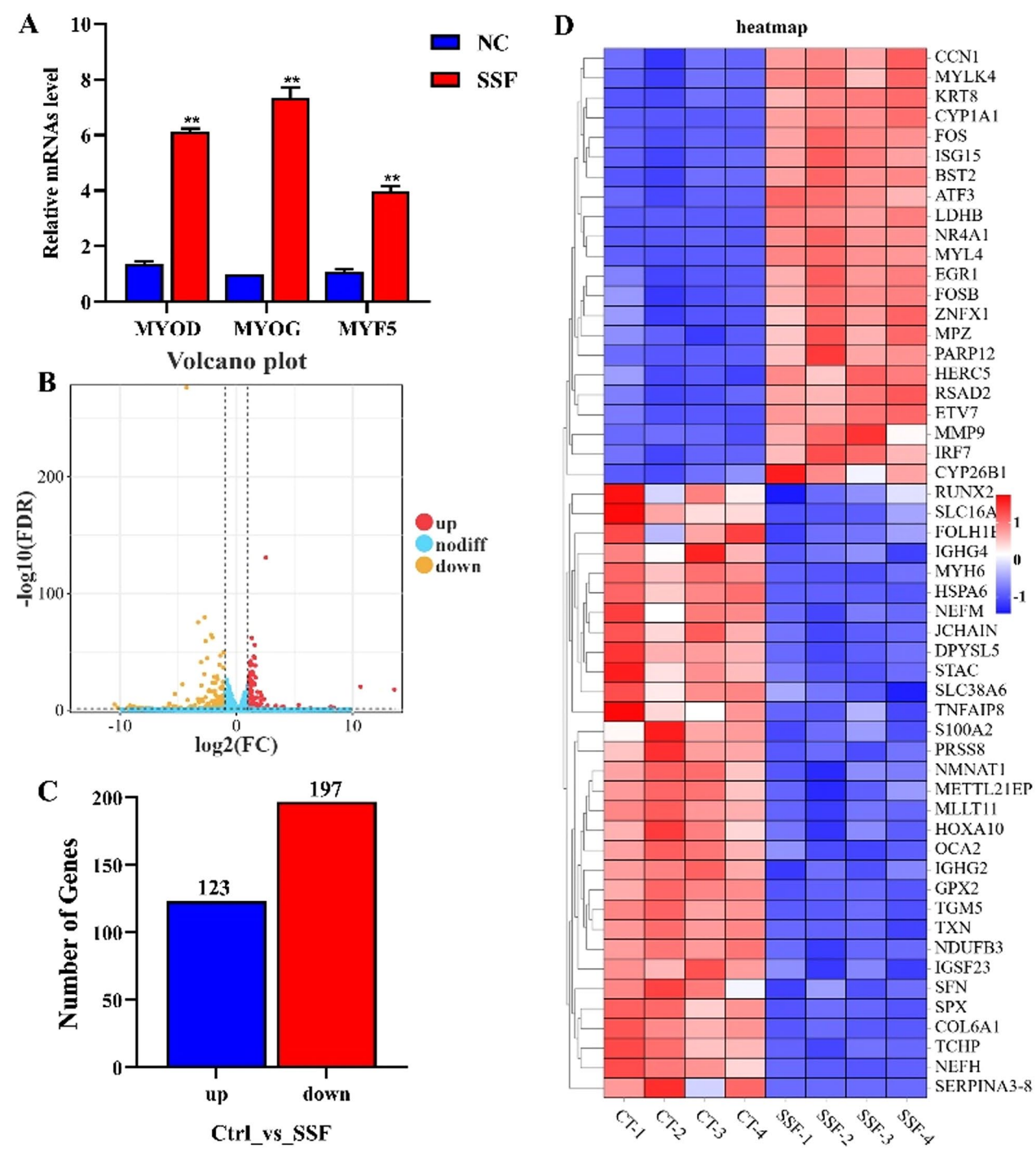


Fig. 8 Analysis of differential genes associated with muscle development. **A** Expression of muscle development-related factors between the two groups; **B** Volcano plot and **C** Bar chart of differentially expressed genes; **D** heatmap; **E** KEGG pathway analysis related to metabolism

can pose risks to animal health [29]. Among these, TI in ANFs can hinder digestion [30], and PHA can damage pig intestinal tissue [31]. Consistent with previous research findings [32], this study discovered that after fermentation, α -conglycinin, β -conglycinin, SPag, TI, and PHA were significantly reduced, and the contents of toxins DON and ZEN were significantly decreased compared to

the ctrl group. These results indicate that SSF can effectively degrade ANFs in feed, has a positive effect on feed quality improvement, and can more effectively reduce the risk of vomiting and poisoning in growing pigs caused by toxins.

During the fermentation process of feed, the involvement of microorganisms promotes changes in

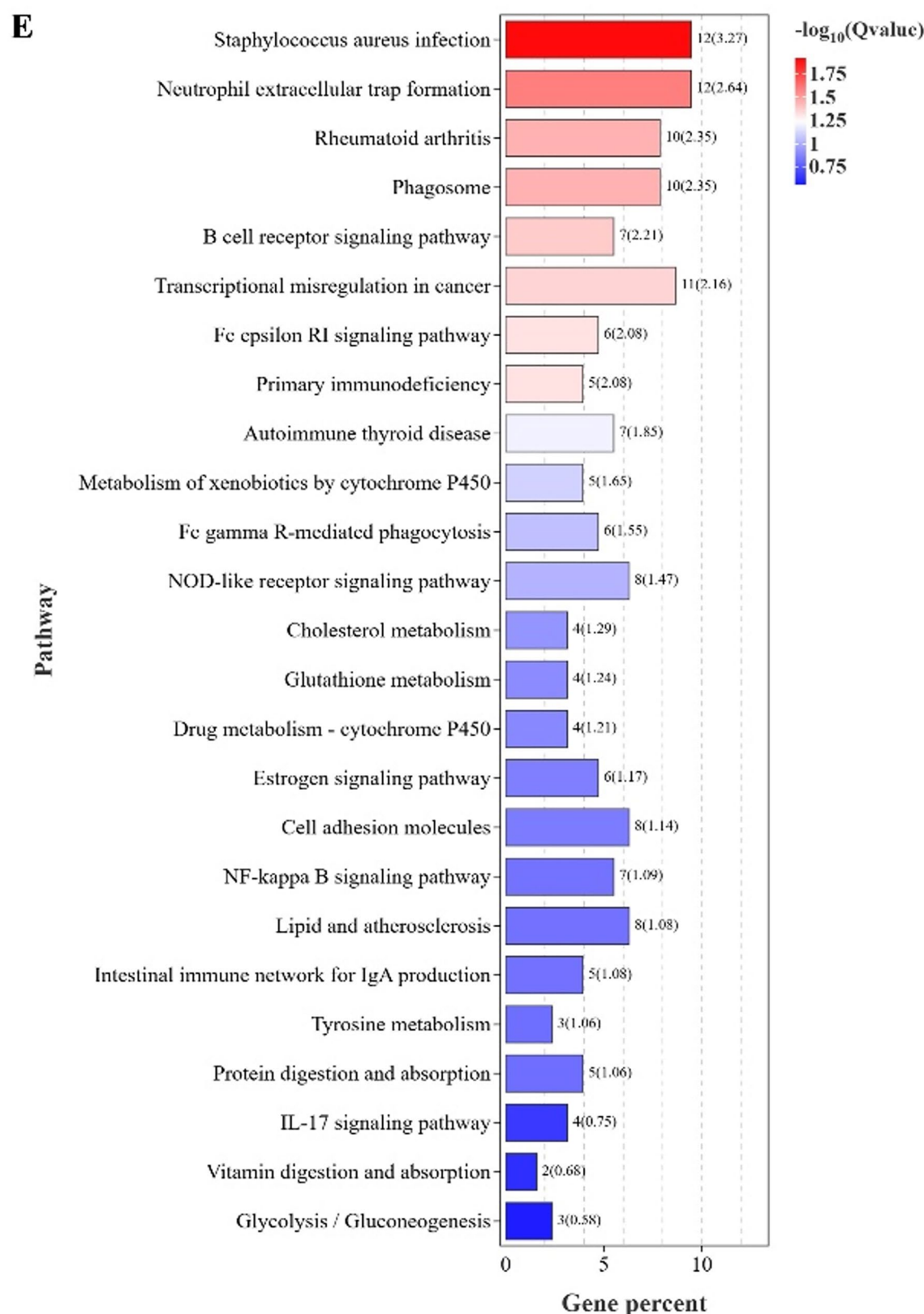


Fig. 8 (continued)

certain metabolites within the feed. This experiment further employed 16 S rDNA sequencing and flavoromics

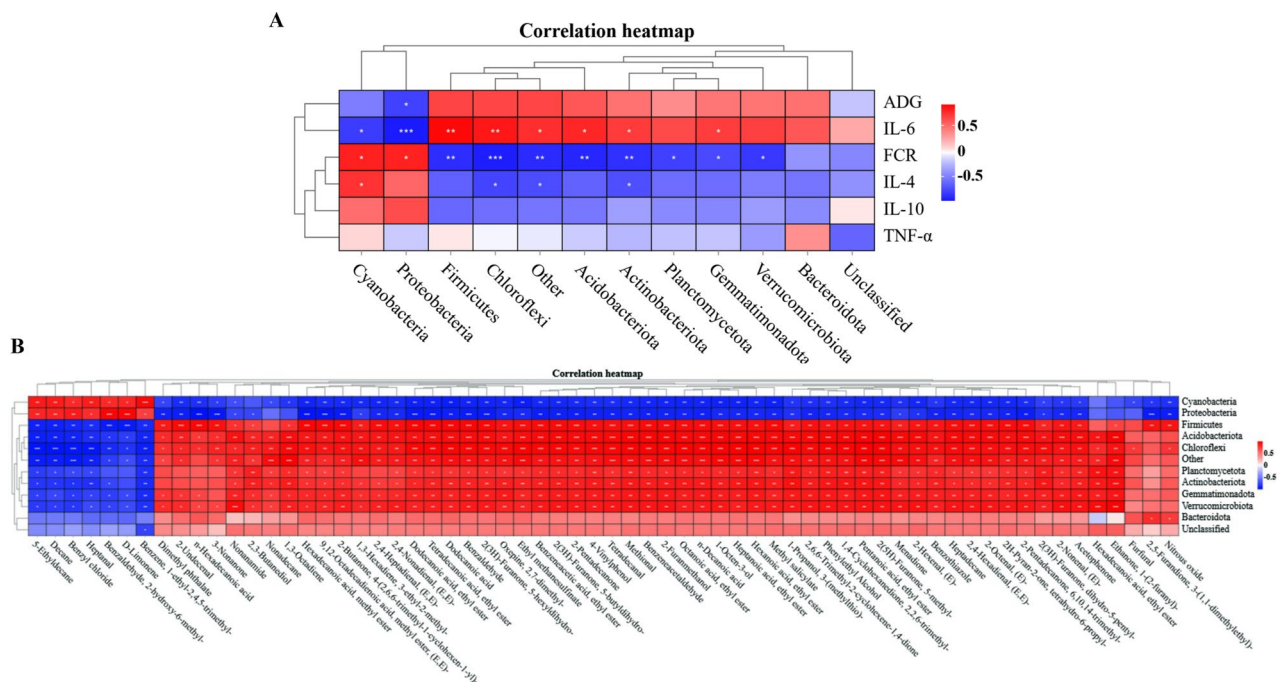


Fig. 9 Construct interaction network diagrams between environmental factors, microorganisms, and metabolites. **A** An interaction network diagram between environmental factors and microorganisms. **B** An interaction network diagram between microorganisms and metabolites

sequencing to investigate the microbial community composition and flavor metabolite in feed. The results showed that, as the fermentation progressed, the phyla *Acidobacteria* and *Firmicutes* significantly increased, and the core bacterial genera shifted from *Pseudomonas* to *Lactobacillus*. This shift may be associated with the acid-producing activity of *Lactobacillus plantarum* [33] and the strong proteolytic and cellulolytic capabilities of *Bacillus* during fermentation [34]. Meanwhile, as feed fermentation progressed, the abundance of genes related to carbohydrate and amino acid metabolism significantly increased, indicating that these metabolic functions may be associated with the production of carbohydrates and membrane transport. Given that current research primarily focuses on the impact of fermented feed on pork flavor [35, 36], while studies on characteristic flavor metabolites of fermented feed are relatively rare. It has been reported that the fermentation of corn flour with *Lactobacillus plantarum* and *Saccharomyces cerevisiae* significantly increases aromatic compounds and alcohol flavor substances, resulting in a strong fruity aroma [37]. In this experiment, the increase in phenolic and heterocyclic compounds can also release a special aromatic odor. However, the trend of changes in alcohol metabolites is opposite, which may be due to the combination of alcohols with some acidic substances in the substrate to produce more ester compounds, thereby reducing their relative content. Interestingly, the key flavor compounds such as 2,3-butanediol, 2-nonanone, Benzyl alcohol, γ -decalactone, and γ -octalactone identified in this study

have also yielded the same results in other studies [38]. It is worth noting that the current experimental design cannot definitively distinguish whether the observed changes in the gut microbiota are attributable to the direct colonization of viable microorganisms carried in the fermented feed or to the synergistic effects of both factors. Further in-depth investigation is required to elucidate this aspect.

Growth performance is the most direct indicator reflecting the feeding effectiveness of fermented feed [39]. It has been reported that fermented feed can significantly increase ADG and decrease FCR, but has no significant effect on ADFI [40], which is consistent with the results of this experiment. Further analysis through interaction plots revealed that the significant increase in *Firmicutes* was positively correlated with the production of SCFA-related metabolites such as 2,3-butanediol and 2-nonanone, which in turn promoted pig growth and improved feed conversion efficiency. Conversely, the reduction in *Cyanobacteria* may be associated with enhanced growth performance in pigs. It was found that in the early developmental stage of growing pigs, SSF significantly improved muscle tenderness, but had no significant effect on meat color and pH value, which this may be related to the early improvement of the myofibrillar protein network structure [41]. Currently, Serum physiological, immune, and antioxidant indicators have been proven to comprehensively reflect the animal's basic nutritional metabolism and immune status [42]. In this study, SSF significantly increased serum IgA and IgG levels, indicating its positive role in enhancing animal

immune function. The study also observed that in the SSF group, the levels of the pro-inflammatory factor IL-6 and the oxidative factor MDA in serum were significantly reduced, while the levels of the antioxidant indicators GSH-PX and SOD were significantly increased, indicating that fermentation effectively enhanced the immune stress resistance of growing pigs. The small intestine is the primary site for nutrient digestion and absorption in pigs and is closely related to growth performance [43]. Studies have shown that the key to nutrient absorption in the small intestine lies in the integrity and morphology of intestinal villi. The higher the villus-crypt ratio, the greater the absorptive surface area and the stronger the absorption capacity of the villi [44]. Moreover, tight junction proteins Claudin-1, ZO-1, and Occludin are important components of the intestinal barrier, selectively permeating nutrients and water between intestinal wall cells and preventing pathogens from entering the body through the intestinal lumen [45, 46]. These results showed that the addition of SSF feed significantly improved the structural morphology of the small intestine and increased the relative expression of Occludin, Claudin-1, and ZO-1 proteins in the jejunum and ileum of growing pigs. Certainly, the relatively low number of replicate pens per treatment group in this study may limit the generalizability of the findings. However, as this was an exploratory study aimed at generating preliminary insights, future research with an expanded sample size is warranted to validate these results.

Conclusion

This study successfully established and optimized a SSF process based on a combination of *B. subtilis*, *L. plantarum*, *S. cerevisiae*, and *A. niger*. This process significantly enhances the nutritional value of feed through the synergistic action of microbes and enzymes, increases the abundance of beneficial bacterial populations, and improves flavor characteristics. After feeding growing pigs a diet containing 10% fermented feed, it effectively improved muscle development and tenderness of the meat, increased serum antioxidant and immune capabilities, significantly improved the morphology of the small intestine tissue, and enhanced the intestinal barrier function, which is beneficial for the nutritional absorption and healthy growth of growing pigs.

Abbreviations

Met	Methionine
Thr	Threonine
Trp	Tryptophan
DM	Dry matter
OM	Organic matter
CP	Crude protein
CF	Crude Fiber
EE	Ether extract
ADF	Acid detergent fiber
NDF	Neutral detergent fiber

CA	Crude ash
Lys	Lysine
Cys	Cystine
B. Subtilis	Bacillus subtilis
L. Plantarum	Lactobacillus plantarum
S. Cerevisiae	Saccharomyces cerevisiae
A. Niger	Aspergillus niger
DON	Deoxynivalenol
ZEN	Zearalenone
AFB1	Aflatoxin B1
TS	T-2 toxin
ADG	Average daily gain
ADGI	Average daily gain intake
FCR	Feed conversion ratio
TP	Total protein
ALB	Albumin
GLOB	globulin
GGT	Glutamyl transferase
ALT	Alanine aminotransferase
Crea	Creatinine
UN	Urea
GLU	Glucose
INS	Insulin
GH	Growth hormone
IL-4	Interleukin-4
IL-6	Interleukin-6
IL-10	Interleukin-10
TNF-α	Tumornecrosis-α
IgA	Immunoglobulin A
IgG	Immunoglobulin G
CAT	Catalase
SOD	Superoxide dismutase
T-AOC	Total antioxidant capacity
GSH-PX	Glutathione peroxidase
MDA	Malondialdehyde
VH	Villus height
CD	Crypt depth

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s42523-026-00540-1>.

Supplementary Material 1

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Author contributions

Mengting Ji: Validation, Formal analysis, Funding acquisition, Writing - Original Draft and Writing - Review & Editing. Jingchao Liu: Validation and Formal analysis. Qinglin Wang: Conceptualization and Formal analysis. Tianye Gong: Data curation, Conceptualization and Methodology. Jiaqi Zhang: Formal analysis, Investigation. Meng Li: Data Curation. Xiaohong Guo: Supervision and Resources. Yang Yang: Supervision, Project administration, Funding acquisition and Writing - Review & Editing. Bugao Li: Project administration, Funding acquisition and Writing - Review & Editing.

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Data availability

The 16S dataset used and analyzed during this study can be found at NCBI SRA under accession number PRJNA1275368, the RNA-seq dataset is also available at NCBI SRA under the same accession number PRJNA1282860, and the metabolomics datasets are available under NGDC OMIX accession OMIX011743. The above data that support the study findings are publicly available. Information can be made available from the authors upon request.

Declarations

Ethics approval and consent to participate

All experimental procedures were approved by the Animal Ethics Committee of Shanxi Agricultural University (approval number: SXAU-EAW-2021MS.P052801). The experimental protocols complied with the requirements for farm animal welfare and were approved by the College of Animal Science and Technology at Shanxi Agricultural University.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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